DIPLOMA IN ELECTRICAL AND ELECTRONICS ENGINEERING • SEMESTER 4

Power Electronics Devices and Circuits

This corrected edition turns 4031 into a proper engineering textbook. It explains device physics only as much as needed for application, then moves into controlled rectifiers, choppers, inverters, SMPS, UPS and drives with diagrams, waveforms, formulas, numerical problems and field notes.

COURSE CODE 4031	CREDITS 4
PERIODS 60	TYPE Program Core

> Power electronics □□□□□□□□□□□□□ device name □□□□□□□ □□□□□□□□. Symbol, current path, waveform, firing angle, duty cycle, protection, heat sink, load effect □□□□□□□ □□□□□□ □□□□□□□□□□.

COURSE BLUEPRINT

What this handbook is expected to teach

This section converts the dull syllabus table into a usable engineering route: prerequisite knowledge, outcome target, field meaning, diagrams to draw, and problems to solve.

Official objectives

- ✓ Introduce power semiconductor devices and passive parts used in power electronics.
- ✓ Explain operation of controlled rectifiers and converters.
- ✓ Build foundation for further study of power electronics and drives.
- ✓ Connect circuits with industrial power conversion applications.

Prerequisite stack

- ✓ Basics of electronics from Fundamentals of Electrical and Electronics Engineering.
- ✓ Analog and digital electronic circuits.
- ✓ Electrical circuit concepts, AC and DC loads.
- ✓ Waveform, RMS, average value and switching concepts.

How to use

1. Read the concept explanation.
2. Study the diagram or waveform.
3. Solve the worked example without looking.
4. Write the expected-answer format.

CO	Outcome	Hours	Engineering meaning
CO 1	Select appropriate power semiconductor device for an application.	16	Choose MOSFET, IGBT, SCR, TRIAC or related device based on voltage, current, switching speed and control need.
CO 2	Summarize principle and operation of controlled rectifiers.	14	Explain firing angle, load current path, waveform and freewheeling action.
CO 3	Summarize DC and AC converters.	12	Explain choppers, buck, boost and cycloconverter operation.
CO 4	Summarize inverters and electric drives.	16	Explain bridge inverters, PWM, SMPS, UPS and drives.

REDESIGNED SYLLABUS STRUCTURE

Module map with diagrams, applications and problem targets

The old plain syllabus table is not enough. This map tells what to draw, what to explain, where the marks come from, and how the concept is used in industry.

Module	Core topics	Must draw / visualize	Must solve / apply	Hours
M1: Power semiconductor devices and SCR control	MOSFET, IGBT, GTO, LASCR, SCS, UJT, DIAC, TRIAC, SCR, turn-on, turn-off, snubber and datasheet reading.	Symbols, SCR V-I curve, PNPN structure, two transistor analogy and snubber idea.	Device selection, holding/latching current, dv/dt and datasheet parameter questions.	16
M2: Controlled rectifiers and firing angle control	Half wave, full wave, semi-controlled and three-phase controlled rectifiers with R and RL loads.	Half wave rectifier, bridge rectifier, RL load with freewheeling diode, waveforms.	Average output voltage and firing angle trend problems.	14
M3: Choppers, buck boost converters and cycloconverters	DC chopper principle, control strategies, Type A to E choppers, buck, boost and cycloconverters.	Chopper waveform, Type A to E quadrant idea, buck and boost diagram, cycloconverter block.	Duty cycle and output voltage problems.	12
M4: Inverters, SMPS, UPS and electric drives	Series inverter, parallel inverter, VSI, CSI, bridge inverter, PWM, SMPS, UPS and electric drives.	Series inverter, full bridge inverter, three-phase bridge, PWM waveform, SMPS block, UPS block and drive block.	PWM concept, inverter application and UPS comparison questions.	16

Power semiconductor devices and SCR control

Power electronic devices are high-energy switches. The exam asks symbols and characteristics, but engineering use demands gate drive, thermal design and protection understanding.

Textbook explanation

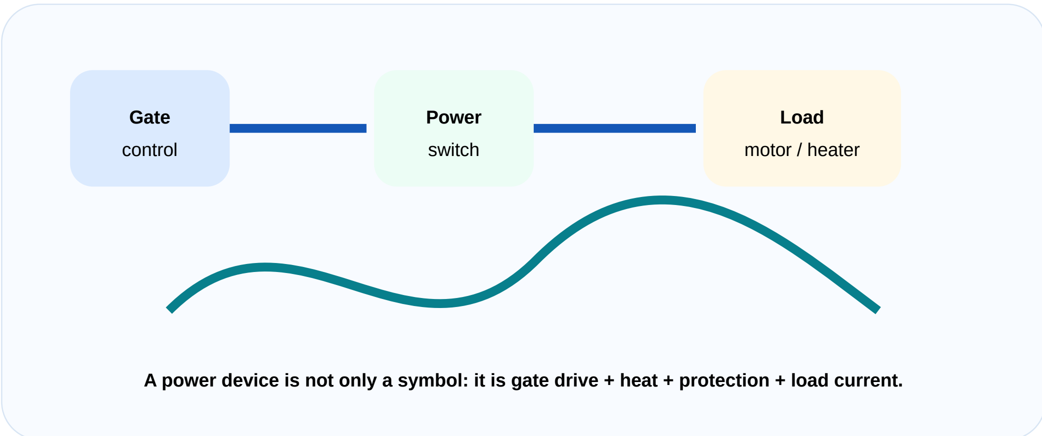
MOSFET is useful for high-speed low-to-medium voltage switching. IGBT is widely used for high-power inverter and drive applications.

SCR has PNPN structure. It can be turned ON by gate pulse but it remains latched until current falls below holding current.

UJT can be used for triggering pulses. DIAC and TRIAC are important for AC control.

Datasheet interpretation includes voltage rating, current rating, gate trigger current, holding current, dv/dt and thermal limits.

> SCR gate current ON □□□□□□□□ □□□□□□□□□□□, OFF □□□□□□□□□□. Datasheet rating ignore □□□□□□□□ device burn □□□□.



Power device as gate drive plus protected load current path.

Engineering subtopics

Device symbols

Know symbol and terminal names.

- ✓ MOSFET gate, drain, source.
- ✓ IGBT gate, collector, emitter.
- ✓ SCR anode, cathode, gate.

SCR V-I curve

Forward blocking, conduction and reverse blocking.

- ✓ Breakover voltage.
- ✓ Holding current.
- ✓ Latching current.

Turn-on methods

Gate, light, dv/dt, thermal and forward voltage.

- ✓ Gate triggering is controlled method.
- ✓ dv/dt triggering is usually unwanted.

Turn-off methods

Natural and forced commutation.

- ✓ AC circuits use natural commutation.
- ✓ DC circuits need forced commutation.

Snubber

Protection against dv/dt and voltage spikes.

- ✓ RC snubber across SCR.
- ✓ Protects against false triggering.

Datasheet reading

Device selection from ratings.

- ✓ Voltage margin.
- ✓ Current and heat sink.
- ✓ Gate trigger rating.

Important formulas / rules

MOSFET loss: $P = I^2 R_{DS(on)}$

SCR OFF condition: $I_A < I_H$ for required time.

Thermal safety depends on power loss and heat sinking.

Common mistakes

- ✓ Memorizing device names without terminal functions.
- ✓ Not mentioning snubber in protection.
- ✓ Thinking holding current and latching current are same.
- ✓ Selecting device without checking voltage and current rating.

Worked examples inside this module

Choose device for high-power inverter drive.

IGBT is suitable because it handles high voltage/current and has easier voltage-controlled gate drive.

Why snubber is used across SCR?

Snubber limits dv/dt and transient voltage, reducing false triggering and device stress.

Controlled rectifiers and firing angle control

Controlled rectifiers are the standard way to convert AC into adjustable DC using SCR firing angle control.

Textbook explanation

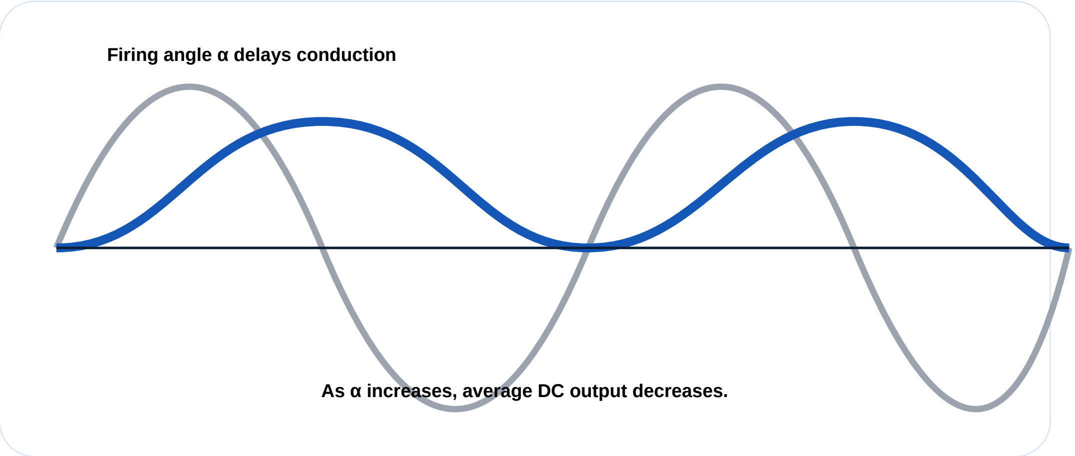
Firing angle is the delay between natural zero crossing and SCR turn-on. As firing angle increases, average DC output decreases.

With RL load, current continues after source voltage becomes zero due to energy stored in inductance. Freewheeling diode improves current continuity.

Full-wave controlled rectifier gives better output and lower ripple than half-wave. Semi-controlled converters use both SCRs and diodes.

Three-phase bridge converter gives smoother output and is used in higher power systems.

> Rectifier answer-□ R load, RL load, freewheeling diode, waveform difference □□□□□□□ clear □□□□□□□.



Firing angle controls controlled rectifier output.

Engineering subtopics

Firing angle

Delay angle of SCR triggering.

- ✓ Controls average output.
- ✓ Measured from zero crossing.

Half-wave controlled rectifier

One SCR controls one half cycle.

- ✓ Simple low power circuit.
- ✓ High ripple.

Full-wave controlled rectifier

Uses both half cycles.

- ✓ Bridge or centre tapped.
- ✓ Better average output.

RL load

Current lags and continues.

- ✓ Inductance stores energy.
- ✓ Freewheeling diode helps.

Semi-controlled rectifier

SCR and diode combination.

- ✓ Partial control.
- ✓ Common in medium power.

Three-phase bridge

Higher power rectifier.

- ✓ Smoother DC.
- ✓ Used in drives.

Important formulas / rules

Half-wave controlled rectifier: $V_{avg} = (V_m/2\pi)(1+\cos\alpha)$

Full converter continuous mode: $V_{avg} = (2V_m/\pi)\cos\alpha$

Conduction angle decreases as firing angle increases.

Common mistakes

- ✓ Forgetting to label firing angle α.
- ✓ Drawing diode rectifier when SCR rectifier is asked.
- ✓ Ignoring RL load current continuation.
- ✓ Not explaining freewheeling diode.

Worked examples inside this module

For a full converter, what happens when α changes from 30° to 75°?

cos α decreases, so average output voltage decreases.

Why freewheeling diode is added for RL load?

It gives path for inductive load current and reduces negative voltage and ripple.

Choppers, buck boost converters and cycloconverters

Choppers and converters use high-speed switching to control average voltage and power flow.

Textbook explanation

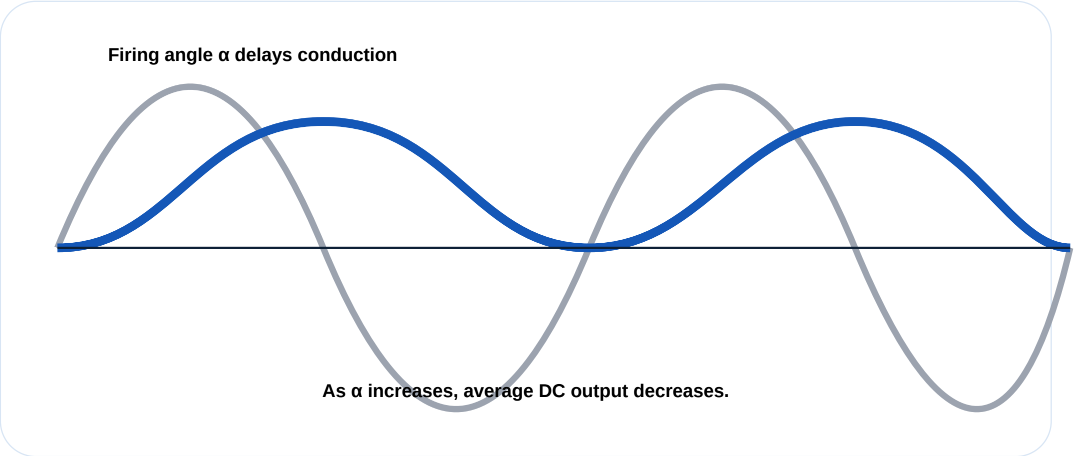
A DC chopper converts fixed DC into variable DC. Duty cycle controls average output voltage.

Constant frequency control changes ON time at fixed switching frequency. Variable frequency control changes switching frequency.

Buck converter steps voltage down. Boost converter steps voltage up.

Cycloconverter converts AC at one frequency directly to AC at another frequency using controlled rectifier groups.

> Duty cycle formula □□□□□□□□ apply □□□□□□□□. Step-down and step-up chopper confuse □□□□□□□□□□.



Switching waveform and average output control.

Engineering subtopics

DC chopper

DC to variable DC converter.

- ✓ High-speed switch.
- ✓ Average output control.

Control strategies

Constant and variable frequency control.

- ✓ PWM control.
- ✓ Frequency modulation control.

Quadrants

Type A to Type E choppers.

- ✓ Power flow direction.
- ✓ Motoring and braking regions.

Buck converter

Step-down DC converter.

- ✓ Output below input.
- ✓ Inductor smooths current.

Boost converter

Step-up DC converter.

- ✓ Output above input.
- ✓ Energy stored in inductor.

Cycloconverter

AC frequency converter.

- ✓ No DC link.
- ✓ Low frequency high power use.

Important formulas / rules

Duty cycle: $D = T_{ON}/T$

Step-down chopper: $V_o = D\ V_s$

Ideal boost converter: $V_o = V_s/(1\text{-}D)$

Common mistakes

- ✓ Using AC formulas for chopper.
- ✓ Forgetting D must be less than 1.
- ✓ Not explaining quadrant operation.
- ✓ Confusing inverter and cycloconverter.

Worked examples inside this module

A 100 V source uses duty cycle 0.6 in a step-down chopper. Find output.

$V_o = 0.6 \times 100 = 60$ V.

Why boost output is higher than input?

Inductor stores energy during ON period and releases it in series with supply during OFF period.

Inverters, SMPS, UPS and electric drives

Inverters convert DC into AC. In industry, inverters are not isolated topics; they are part of drives, UPS, SMPS and renewable power systems.

Textbook explanation

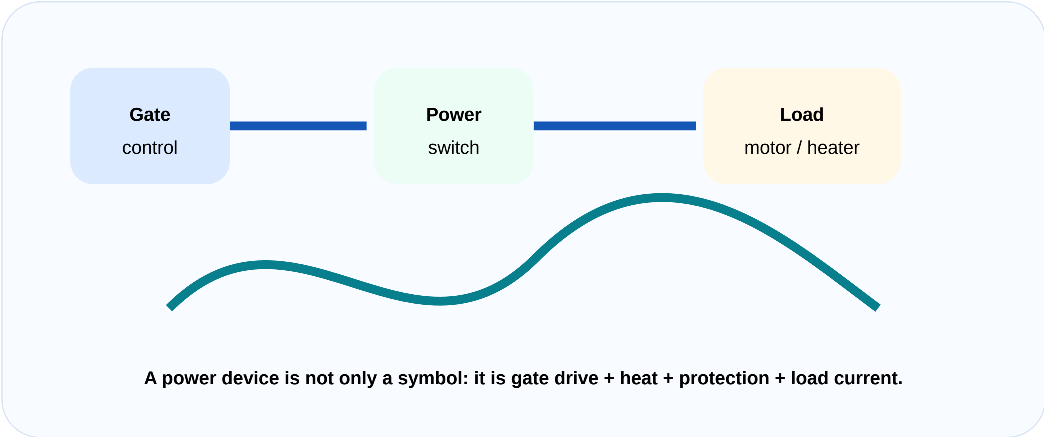
Series and parallel inverters show commutation and oscillation principles. VSI and CSI classify the input source nature.

Bridge inverters use controlled switches. Full bridge gives bipolar output and three-phase bridge feeds AC motors.

PWM controls output voltage and reduces harmonics compared with square wave control.

SMPS converts power efficiently using high-frequency switching. UPS provides backup power through rectifier, battery and inverter stages.

> Inverter answer-□ DC input, switching sequence, output waveform, feedback diode, load type □□□□□□□ mention □□□□□□□□.



DC to AC power conversion for drives and UPS.

Engineering subtopics

Series inverter

Uses resonant load for commutation.

- ✓ Basic inverter type.
- ✓ Used for concept learning.

Bridge inverter

Switching bridge produces AC output.

- ✓ Half bridge.
- ✓ Full bridge.
- ✓ Three-phase bridge.

PWM

Pulse width controls fundamental output.

- ✓ Single PWM.
- ✓ Sinusoidal PWM.
- ✓ Reduces harmonics.

SMPS

High frequency switching supply.

- ✓ Efficient and compact.
- ✓ Needs feedback control.

UPS

Backup supply system.

- ✓ Offline.
- ✓ Online.
- ✓ Line interactive.

Electric drives

Controlled motor system.

- ✓ Converter plus motor plus feedback.
- ✓ Used in pumps, lifts, conveyors.

Important formulas / rules

Inverter output is controlled by switching pattern and DC bus voltage.

PWM duty changes average or fundamental output voltage.

Drive block: source → converter → motor → load → feedback.

Common mistakes

- ✓ Calling inverter AC to DC converter.
- ✓ Not drawing feedback diodes in bridge inverter explanation.
- ✓ Ignoring load type in waveform.
- ✓ Confusing SMPS with linear regulator.

Worked examples inside this module

Why PWM is used in inverter?

PWM controls output voltage and improves waveform quality by reducing low-order harmonics.

Compare online and offline UPS.

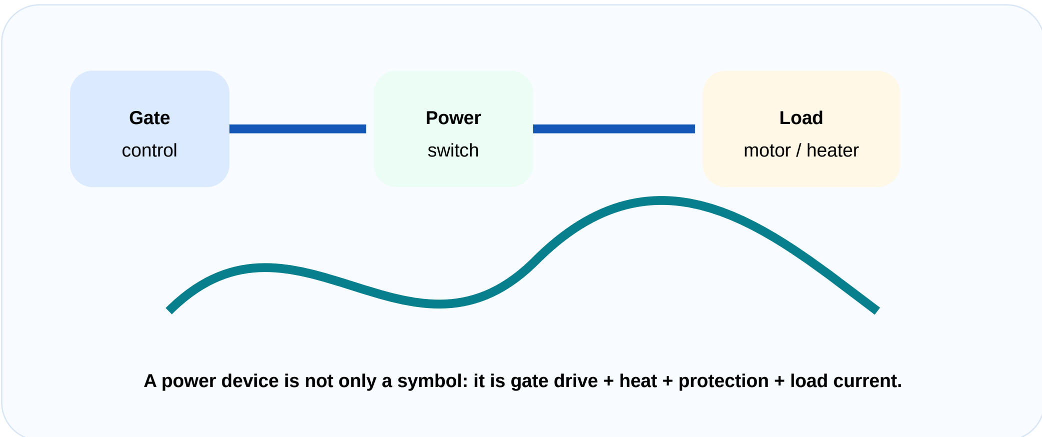
Online UPS always feeds load through inverter with no transfer delay. Offline UPS feeds load from mains normally and switches to inverter during failure.

ILLUSTRATION LAB

Images, block diagrams and waveform thinking

For technical subjects, the diagram is half of the answer. These visuals make the lesson look and work like a textbook instead of a plain note.

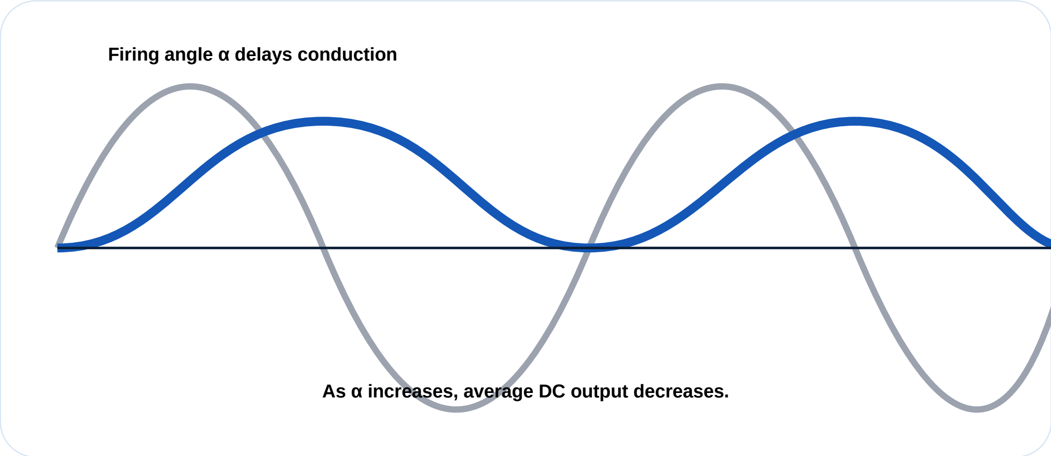
Power device chain



Gate signal controls large power flow.

Use this for device application answers.

Controlled rectifier waveform



Firing angle delay changes average output.

Best visual for rectifier questions.

Converter thinking

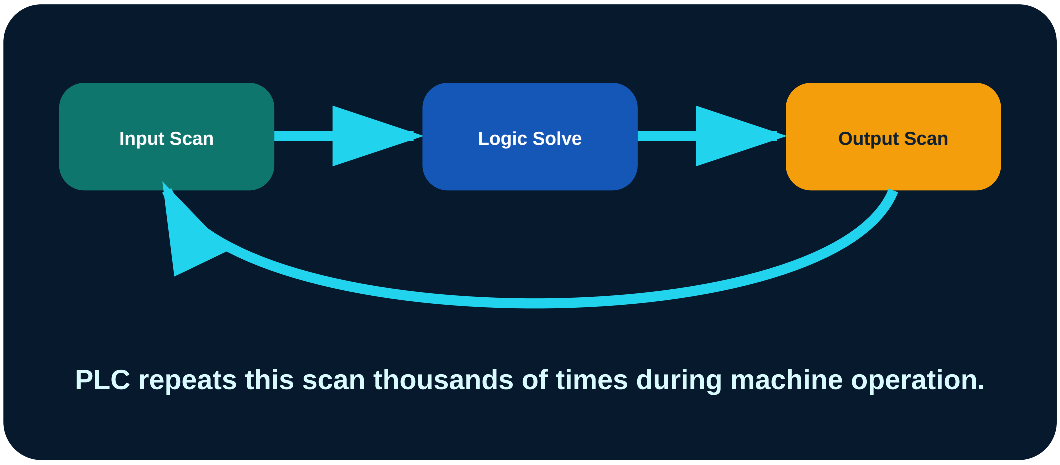
Input power is reshaped to suit the load.

Diagram, waveform, block flow and practical meaning should be studied together.

Input power is reshaped to suit the load.

Power electronics is energy conversion, not only electronics.

Industrial drive block



Controller, converter, motor and feedback form a drive system.

This helps CO4 drive and UPS explanation.

PROBLEM LAB

Solved numerical and design problems

Each problem shows the method, not only the answer. This is the difference between a notebook and an engineering handbook.

Step-down chopper

Problem: A 200 V DC supply feeds chopper with D = 0.25. Find output voltage.

1. Identify as step-down chopper.
2. Use $V_o = D\,V_s$.
3. Substitute 0.25 and 200 V.

Answer: 50 V

Boost converter

Problem: For ideal boost converter Vs = 24 V and D = 0.5. Find output voltage.

1. Use $V_o = V_s/(1-D)$.
2. Substitute 24/(1-0.5).

Answer: 48 V

Controlled rectifier trend

Problem: In full converter, α is increased. What happens to output?

1. Formula contains cos α.
2. cos α decreases as α increases from 0 to 90 degrees.
3. Average output decreases.

Answer: Output voltage decreases.

Device selection

Problem: Choose device for high speed low voltage SMPS.

1. SMPS needs high switching speed.
2. MOSFET has fast majority carrier switching.

Answer: Power MOSFET

PRACTICE

Expected questions by module

Module 1

- List power electronic devices and draw symbols.
- Explain MOSFET and IGBT operation.
- Explain SCR structure and V-I characteristics.
- Classify SCR turn-on and turn-off methods.
- Explain snubber protection.
- Explain SCR datasheet parameters.

Module 2

- Explain single-phase half-wave controlled rectifier.
- Explain full-wave bridge controlled rectifier.
- Explain effect of RL load and freewheeling diode.
- Explain semi-controlled rectifier.
- Outline three-phase controlled rectifier.

Module 3

- Explain DC chopper principle and waveforms.
- Explain control strategies of chopper.
- Classify Type A to E choppers.
- Explain buck and boost converters.
- Explain single-phase cycloconverter.

Module 4

- Explain basic inverter operation.
- Explain bridge inverter with waveforms.
- Explain three-phase bridge inverter in 180 degree mode.
- Explain PWM voltage control.
- Explain SMPS and servo stabilizer.
- Explain UPS and electric drives.

MODEL PAPER

Full practice question paper and answer guide

This model follows the syllabus order and avoids copying official papers.

COURSE 4031 POWER ELECTRONICS DEVICES AND CIRCUITS
Time: 3 Hours   Maximum Marks: 100

PART A
1. Define holding current.
2. State two applications of MOSFET.
3. What is snubber circuit?
4. Define firing angle.
5. What is freewheeling diode?
6. Define duty cycle.
7. What is VSI?
8. What is PWM?
9. State two UPS types.
10. List two electric drive applications.

PART B
11. Explain SCR V-I characteristics.
12. Compare MOSFET and IGBT.
13. Explain SCR triggering methods.
14. Explain half wave controlled rectifier.
15. Explain chopper control strategies.
16. Explain buck and boost converters.
17. Explain bridge inverter.

18. Explain SMPS block diagram.

PART C
19. Explain power semiconductor devices with symbols and applications.
20. Explain controlled rectifiers with R and RL loads.
21. Explain DC choppers and quadrant operation.
22. Explain cycloconverter.
23. Explain PWM inverter and three-phase bridge inverter.
24. Explain online, offline and line interactive UPS.
25. Explain electric drive block diagram.
26. Solve chopper and converter numerical problems.

Answer writing guide

- ✓ Draw symbols and circuits neatly.
- ✓ For waveform answers mark firing angle, conduction interval and load current.
- ✓ For numerical problems write formula, substitution and unit.
- ✓ For applications mention merits, limitations and field use.

REFERENCES

Books and resources

Type	Reference
T1	P S Bimbhra, Power Electronics
T2	M D Singh and K B Khanchandani, Power Electronics
R1	Muhammad H Rashid, Power Electronics Circuits Devices and Applications
R2	Sen P C, Power Electronics
Online	NPTEL power electronics courses, Electrical4U, Swayam and IEEE resources

Final revision rule: Prepare one diagram, one formula, one application and one common mistake for every major topic.